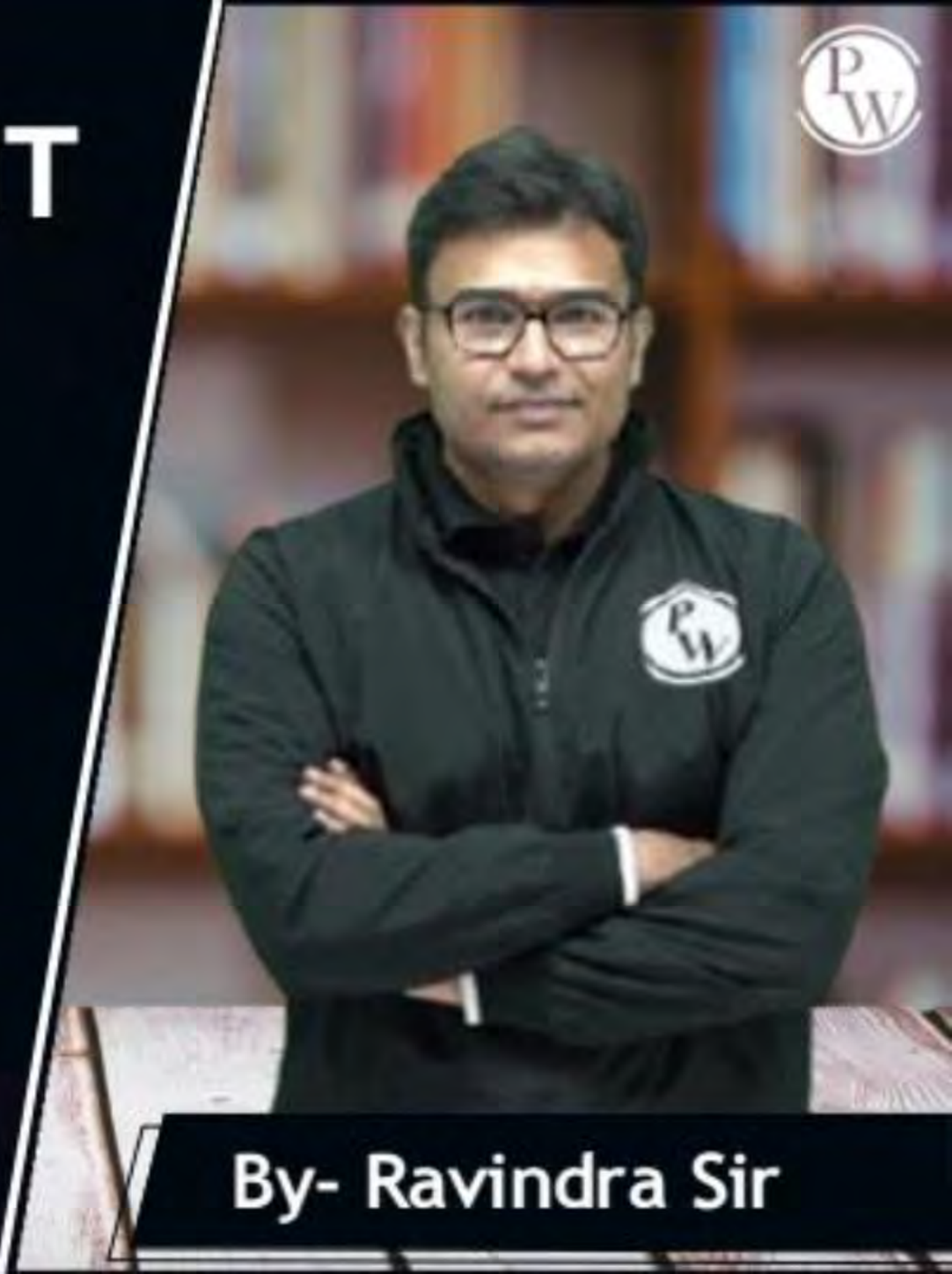
Computer Science & IT

COMPILER DESIGN
(CD)

Code Optimization

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

ICG

Topics to be Covered



- Topic
- Topic
- Topic

CO

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

optimization

m/c independent

machine dependent

machine idependent optimization:-

1) loop optimizations
   a) Code motion
      or
      frequency reduction
   b) loop unrolling
   c) loop jamming
2) Constant folding
3) Constant propagation
4) Redundancy elimination
5) Strength reduction

machine dependent :

1) Register allocation
2) use of addressing modes
3) Peephole optimization
   a) Redundant load/store
   b) flow of control optimization
   c) Strength reduction
   d) use of m/c idioms

Loop optimizations :-

→ To apply loop optimizations, we must first detect loops

→ To detect loops we use control flow analysis (CFA) using program flow graph (PFG)

→ To find program flow graph, we need basic blocks

→ A Basic block is a sequence of 3 address statements where control enters at the beginning and leaves only at the end without any jumps or halts

Loop optimization

↓

loops

↓

CFA (PFG)

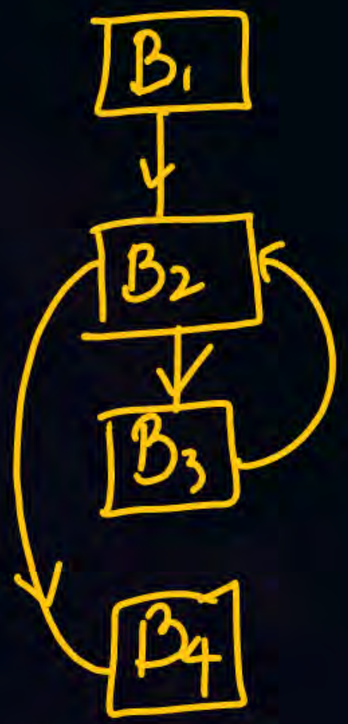
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BB

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Headers

→ Given a program we will construct the program flow graph and examine how the control is flowing



A cycle will indicate a loop.

Finding the Basic blocks :-

In order to find the basic blocks, we need to find the leaders.
Then a BB will start from one leader to the next leader
but not including the next leader.

Identifying a leader in a program:-

1. I statement is a leader
2. Statement that is target of Conditional or unconditional goto statement is a leader
3) Statement that follows immediately a Conditional or unconditional goto statement is a leader.

```
fact(x)
{
  int f=1;
  for(i=2; i<=x; i++)
      f=f*i
  return f;
}
```

If there are n leaders there will be n basic blocks

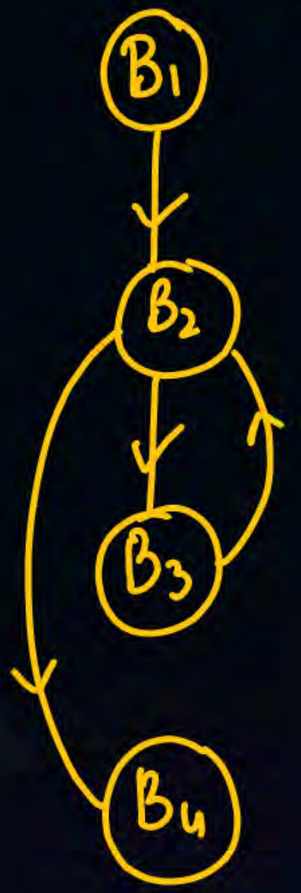
Three address code:-

```
*1. f=1;
 2. i=2;                    B1

*3. if(i>x) goto 9;         B2

*4. t1=f*i;
 5. f=t1;
 6. t2=i+1;                 B3
 7. i=t2;
 8. goto 3;

*9. Goto calling program    B4
```



THANK - YOU